PERSONALIZED AI COMPANION

Project Group 2

Student 01 : Denise Maria James
Student 02 : Alen Augustine Issy
Student 03 : Ann Maria E C
Student 04 : Helan Kuriakose

Under the Guidance of
Miss Anit Thomas M

Dept. of Artificial Intelligence and Data Science
Vimal Jyothi Engineering College, Chemperi

April 18, 2024

Contents

1. Abstract
2. Introduction
3. Objective
4. Scope
5. Problem Definition
6. Literature Survey
7. Implementation And Result
8. Requirement Specification
9. Feasibility Study
10. Conclusion
11. Future Scope
12. References

Abstract

Abstract

This project deals with development of a personalized AI Companion using Conversational AI. Integration with NLP, Open AI, and LLM Agent to provide an all-in-one mental health well being system for everyone.

Introduction

- As Personalized AI Companion,we propose a fully automated and powerful first-level detection and support system for mental health. this project proposes LangChain Chain as our AI Companion agent, and OpenAI Functions Agent as Entity Extractor for entity memory.
- There is no doubt that mental health can also be defined as people's emotional, behavioral and cognitive well-being.People become insecure and resistant towards sharing their feelings to others in order to avoid criticism and being judged.
- This study aims to provide efficient and essential care to the people with mental health concerns according to their needs and supplying the basic information regarding mental health problems through various sources.

Objective

- The main objective is to create a Interactive AI Companion which will be a all-in-one mental health support system.
- By leveraging advanced memory bank techniques, we aim to create a reliable and efficient system that enhances AI companion's ability to learn from past conversations, adapting to the user's personality and context providing tailored experiences.
- Personalized AI companions utilize machine learning algorithms,NLP and CNN to analyze user data and behavior.So that beyond emotional aspects, AI companions assist with tasks, reminders, and organization.
- In essence, personalized AI companions bridge the gap between technology and human interaction, providing support, companionship, and customized experiences.

Scope

- Mentorship and Guidance Support.
- Easily available Emotional Support and Companionship .
- Entertainment and Creativity Platform.
- Productivity Enhancement System.

Problem Definition

- Nowadays,consulting a therapist is a complicated process. So that's where our chatbot comes into play, with the help of AI Companion the people can openly share their medical issues with the chatbot because of its non-judgmental quality.
- The system combines a LangChain LLMChain as the AI Companion agent with an OpenAI Functions Agent for entity memory.
- The memory file handles message history and summaries, storing old messages and user data to provide persistent information for the AI agent.

Literature Survey

The papers selected for literature survey:

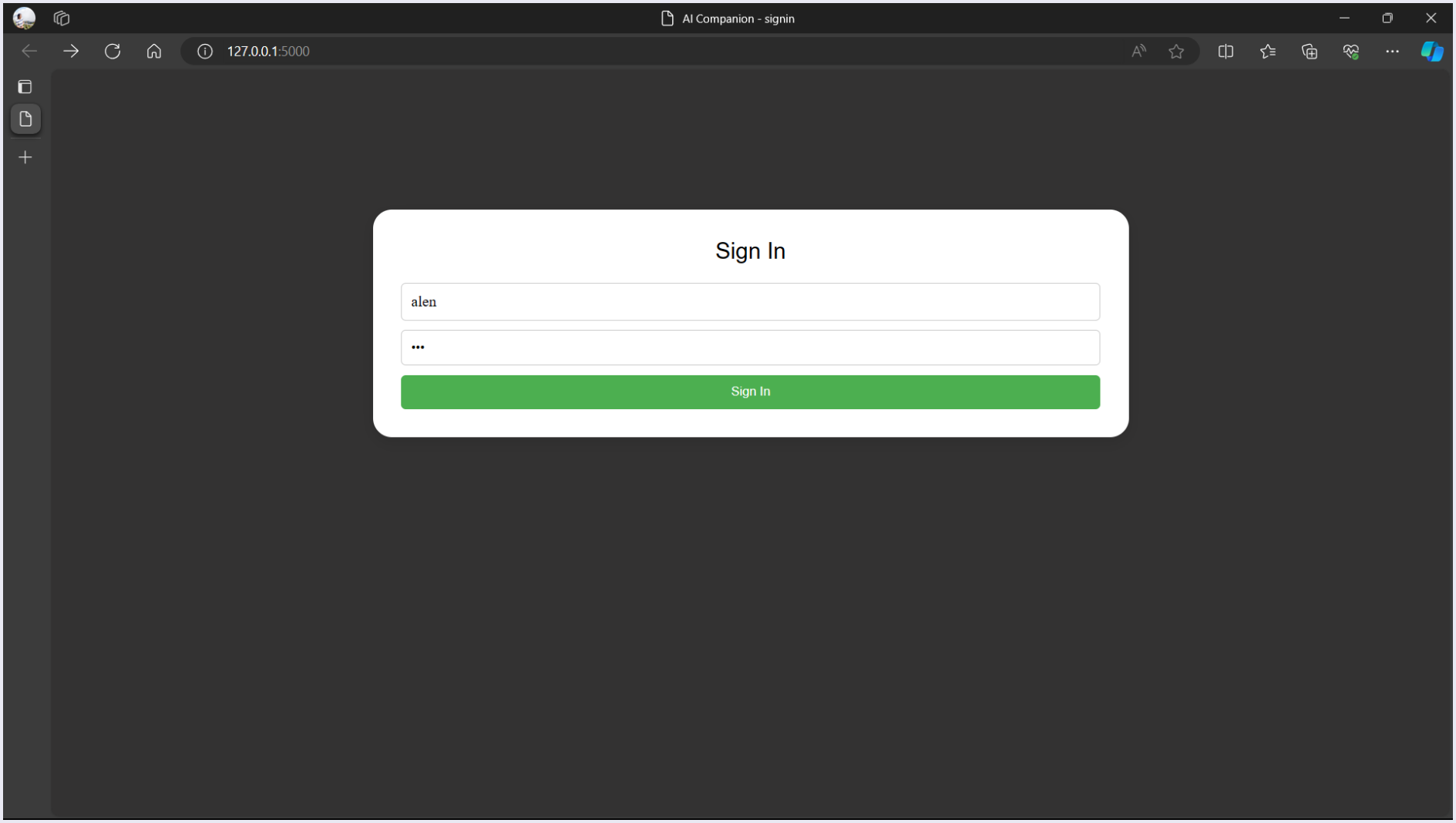
- Preliminary identification and Therapeutic Support of Depression in Mental Health using Conversational AI
- Memory Bank: Enhancing Large Language Models with Long-Term Memory
- The Typing Cure: Experiences with Large Language Model Chatbots for Mental Health Support
- Personalized Response with Generative AI: Improving Customer Interaction with Zero-Shot Learning LLM Chatbots

Consolidated Table

Paper Title	Year	Proposed Solution	Technology used	Accuracy
Preliminary identification and Therapeutic Support of Depression in Mental using Conversational AI	**2023**	**collecting datsets, processing, sentiment analysis, deploying the model**	**NLP, CNN, RNN**	**90%**
MemoryBank: Enhancing Large Language Models with Long-Term Memory	**2023**	**summon relevant memories, continually evolve through continuous memory updates, comprehend, and adapt to a user's personality**	**LLM, Memory Bank**	**91%**
The Typing Cure: Experiences with Large Language Model Chatbots for Mental Health Support	**2024**	**Pre training, Fine tuning Prompt based learning, Large language model**	**LLM**	**92%**
Personalized Response with Generative AI:Improving Customer Interaction with Zero-Shot Learning LLM Chatbots	**2023**	**Train LLM, Input processing, sentiment analysis and conversational style adaption**	**LLM,NLP**	**90%**

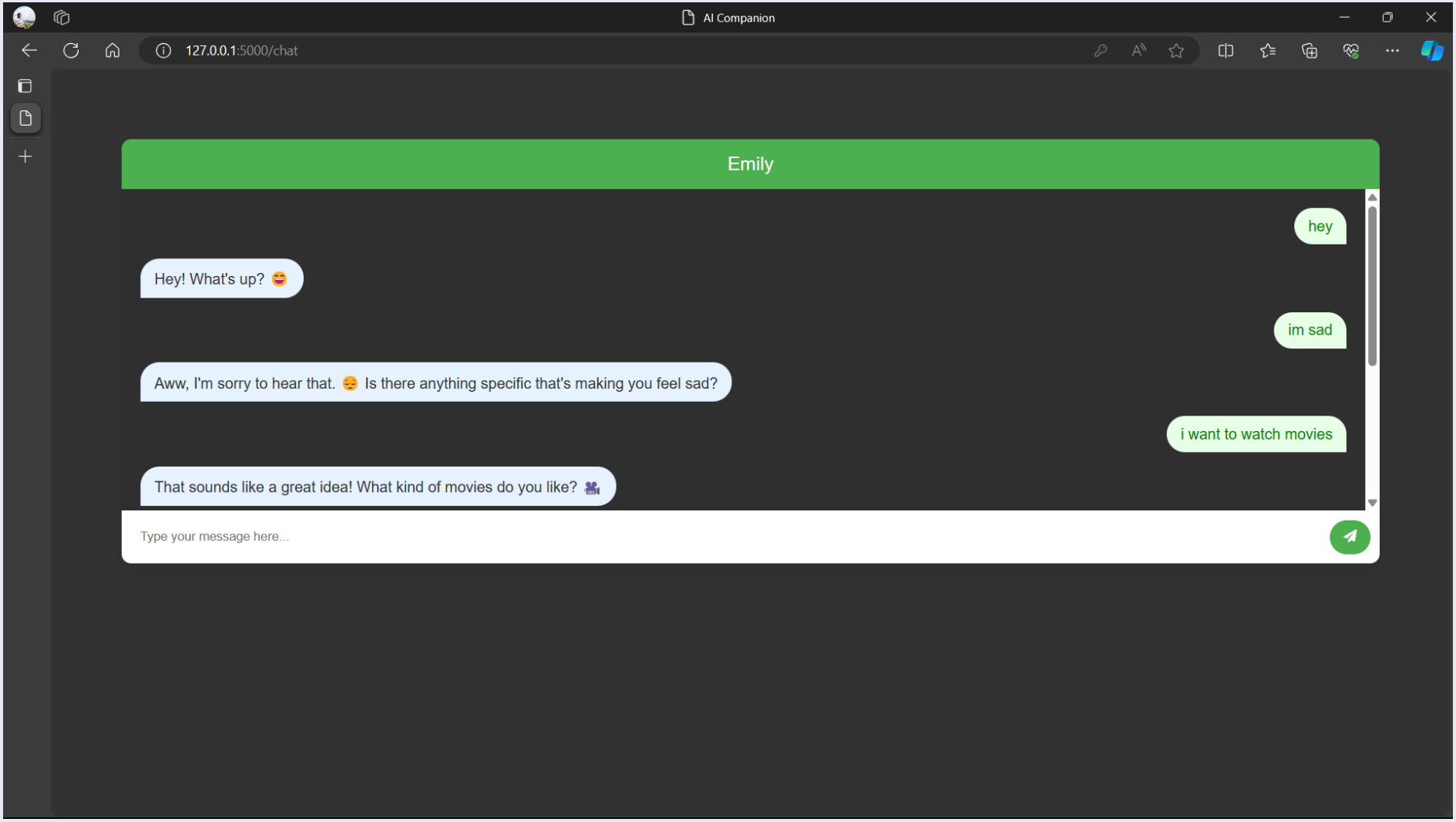
Implementation and Result

Sign-In Page



Implementation and Result

Chat Box



Requirement Specification

User Interface

- Text box for user input.
- Send button for the users to send the message to the model.
- A text area where the send messages and responses are displayed.

Hardware Interface

- No external hardware requirements.

Requirement Specification

Non Functional Requirements

- Empathy
- Speed
- Accuracy
- Reliability

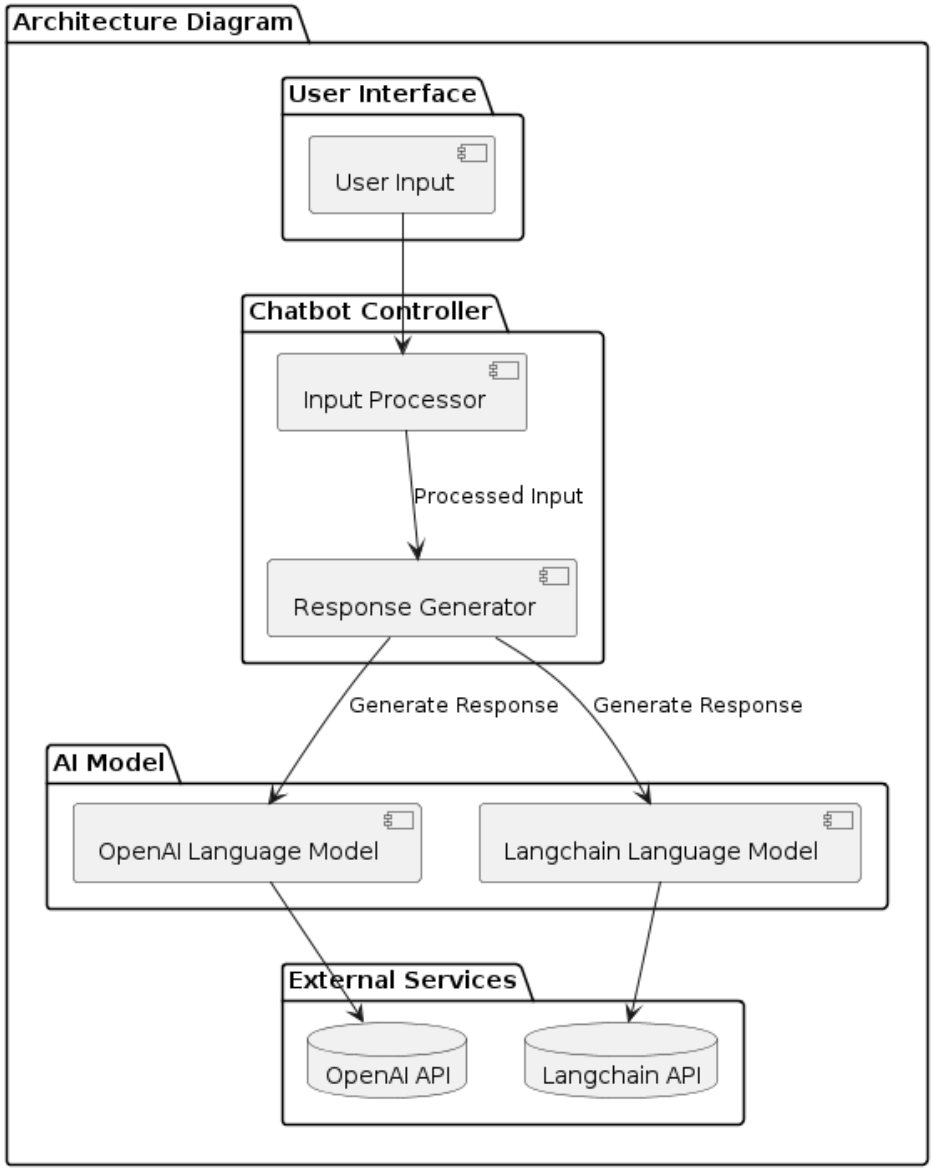
Software Requirements

- Programming Languages: Python
- Libraries used: Lang Chain, Open AI

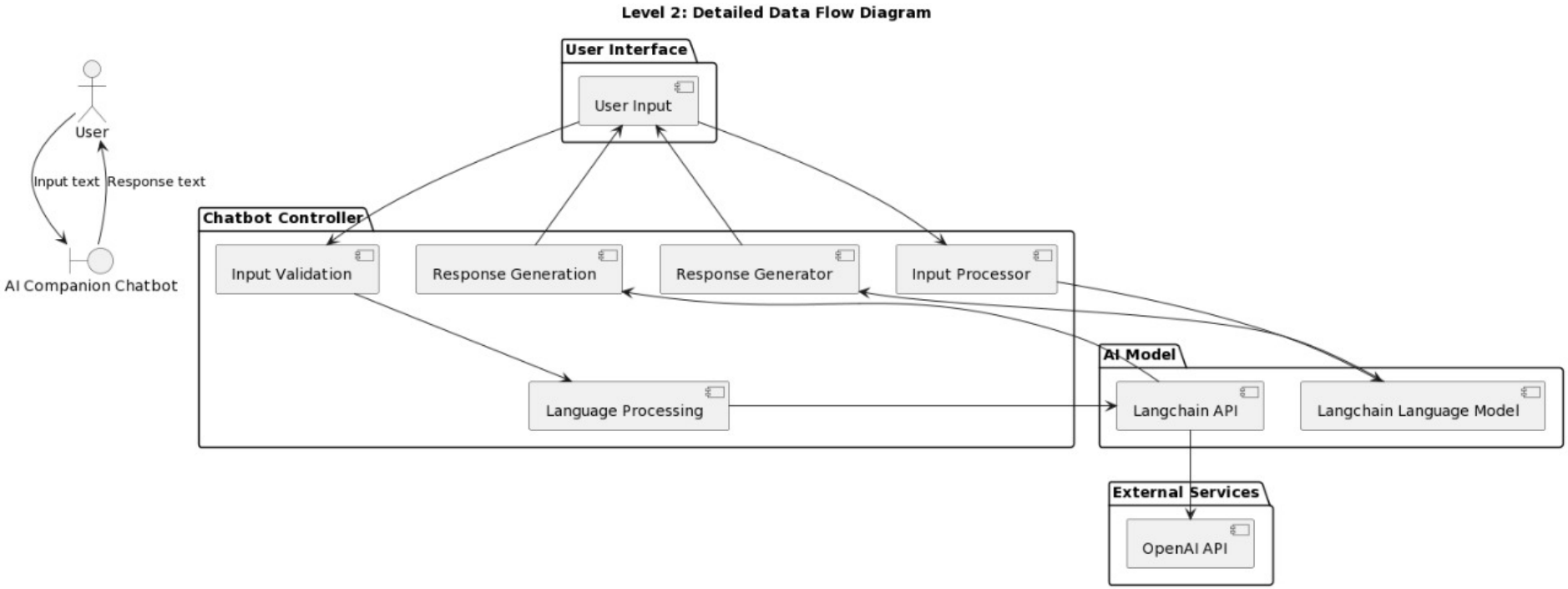
Feasibility Study

- **Technical Feasibility**: The proposed system is technically feasible due to the availability of required hardware and software resources, including advanced deep learning and machine learning algorithms.
- **Operational Feasibility**: The system can be easily integrated into existing software systems with a minimal learning curve for non-technical users.
- **Economic Feasibility**: Although the development cost for the system is minimal, the cost of implementation and maintenance is reasonable.
- **Legal Feasibility**: The legal feasibility of Personalized AI Companion depends on complying with relevant laws and regulations related to data protection, data privacy, and intellectual property rights.

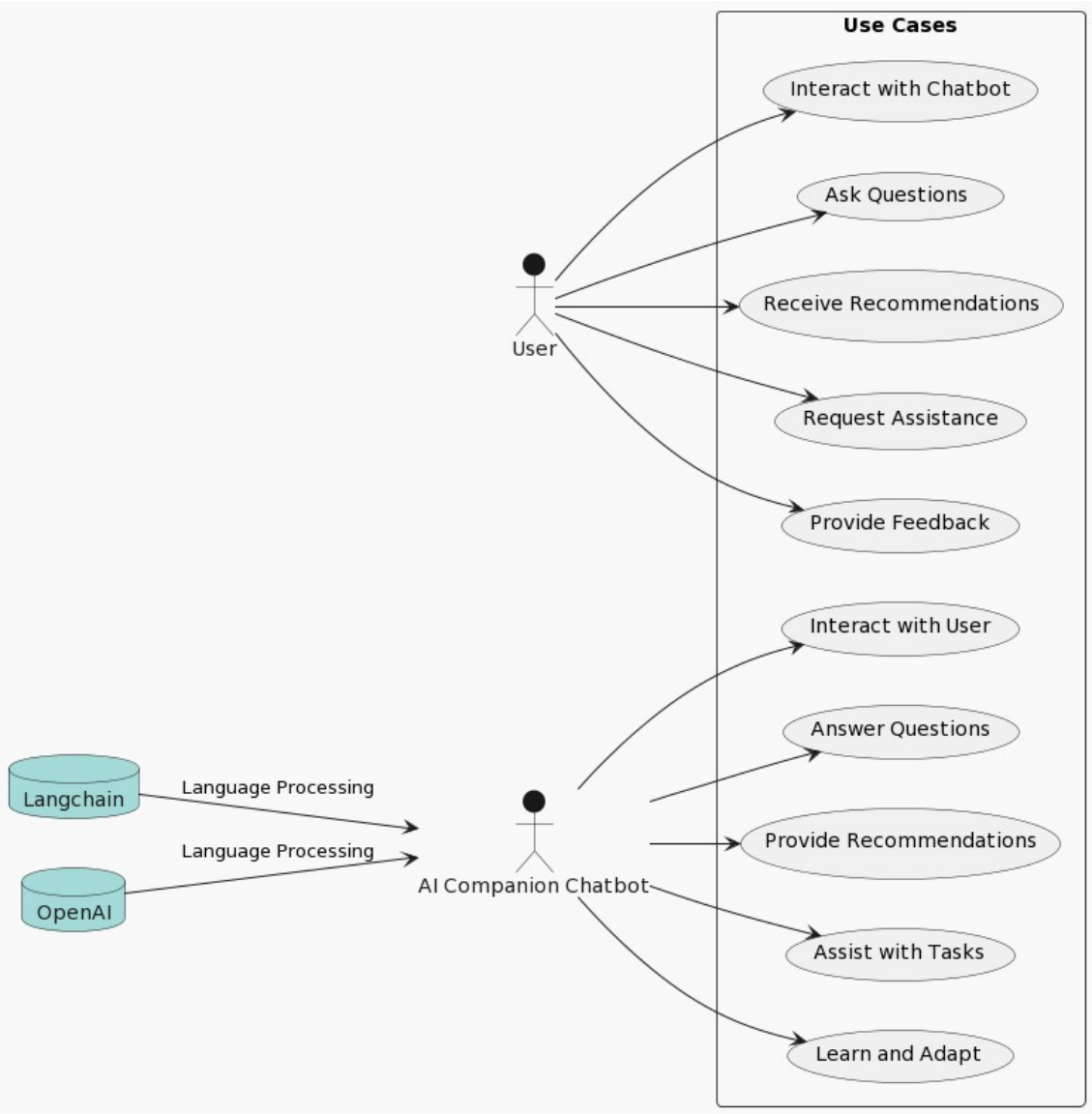
Architecture Diagram



Data Flow Diagram



Use Case Diagram



Conclusion

- The study proposes a all-in-one mental health support system which provides a user-friendly platform to share their thoughts.
- When a user sends a message, Agent 1 responds with a LangChain LLMChain, while Agent 2 determines if it should update the user profile based on the message content.
- This project that involves creating an AI companion similar to Replika or Paradot, which utilizes a custom LangChain Chain as the AI agent and OpenAI Functions Agent for entity extraction to maintain a memory of user interactions.
- This agent provides support system for individuals who are struggling in life.

Future Scope

- Now,to possess a stabilized mental health becomes difficult due to unavoidable situations,Here our chatbot can be a real-life companion.
- It is possible to the extent where we can identify mental health problems by monitoring their response keenly with the help of sensory technology.
- By implementing much more complicated sentimental analysis, we could thoroughly analyze the user's mental health and provide professional support if needed.

References

- Archana Mariappan "Preliminary identification and Therapeutic Support of Depression in Mental Health using Conversational AI" https://knowledgecommons.lakeheadu.ca/handle/2453/5158
- Wanjun Zhong1, Lianghong Guo1, Qiqi Gao2, He Ye3, Yanlin Wang1 "MemoryBank: Enhancing Large Language Models with Long-Term Memory" https://arxiv.org/abs/2305.10250
- INHWA SONG ,SACHIN R. PENDSE, NEHA KUMAR, MUNMUN DE CHOUDHURY "The Typing Cure: Experiences with Large Language Model Chatbots for Mental Health Support" https://arxiv.org/abs/2401.14362
- Joëlle Bink "Personalized Response with Generative AI:Improving Customer Interaction withZero-Shot Learning LLM Chatbots" https://research.tue.nl/en/studentTheses/personalized-response-with-generative-ai

Thank You